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Perioperative dapagliflozin in diabetic patients undergoing cardiac surgery: the DAPA-CS trial

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Introduction

Sodium-glucose cotransporter 2 inhibitors (SGLT2i) have demonstrated profound cardioprotective and renoprotective effects in chronic disease management.[1,2] Theoretically, their pleiotropic effects, including inflammation modulation, oxidative stress reduction, and metabolic reprogramming, may confer perioperative benefits in cardiac surgery.[2] However, whether SGLT2i provide rapid cardiorenal protection in the acute perioperative setting remains unexplored.

Current guidelines recommend discontinuing SGLT2i 3–4 days before surgery to prevent diabetic ketoacidosis (DKA),[3] based on case reports submitted to the FDA rather than systematic perioperative data.[4] However, abrupt withdrawal may lead to adverse outcomes and deprive high-risk patients of potential organ protection.[5] The necessity and timing of perioperative discontinuation remain controversial.[5]

We initiated the DAPA-CS (Dapagliflozin Therapy in Cardiac Surgery) randomized, open-label, blinded-endpoint trial to provide the first evaluation of the efficacy and safety of perioperative dapagliflozin in diabetic patients undergoing on-pump cardiac surgery.

Methods

Participants and intervention

Eligible participants were adults with type 2 diabetes scheduled for elective on-pump cardiac surgery. Key exclusion criteria included current or recent use of SGLT2i, a history of DKA, and severe renal impairment.

Patients were randomized (1:1) to receive dapagliflozin (10 mg/day) plus standard care or standard care alone (without SGLT2i). Dapagliflozin was administered for at least 3 days preoperatively, withheld on the day of surgery, and resumed postoperatively once oral intake was re-established for 5 days. This protocol was designed based on the drug's pharmacokinetics to balance therapeutic presence with safety.[6]

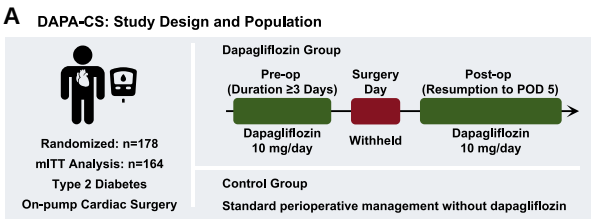
Outcomes

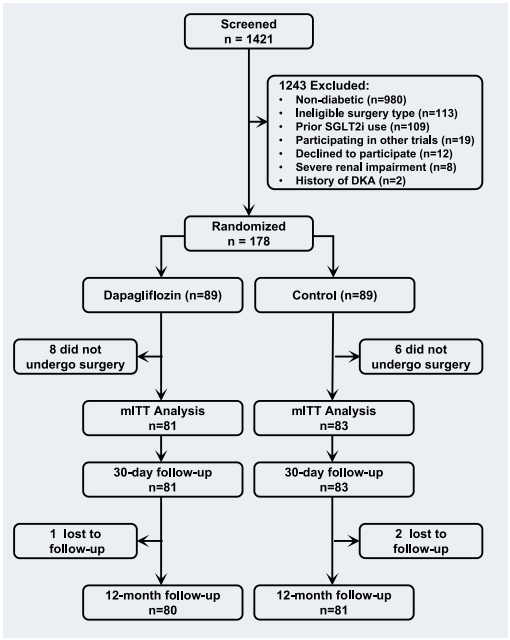
The primary end-point was perioperative myocardial injury, assessed by high-sensitivity cardiac troponin I (hs-cTnI) levels (Abbott ARCHITECT; URL: 0.026 ng/mL) at 48 h postoperatively. The key secondary end-point was the incidence of perioperative acute kidney injury (AKI), defined according to KDIGO criteria.[7] Safety outcomes included DKA, hypoglycaemia, and infections.

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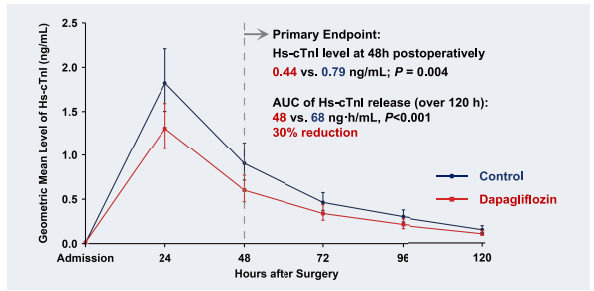
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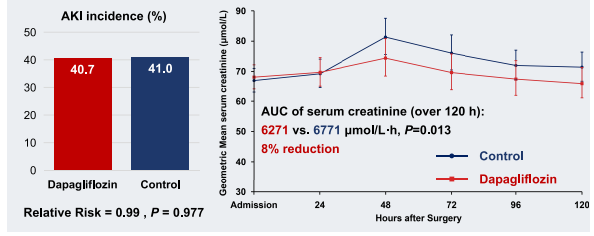




C Baseline characteristics

Characteristic	Overall (n = 164)	Dapagliflozin (n = 81)	Control (n = 83)
Age, years	64 (58-68)	64 (59-67)	65 (57-68)
Men	129 (78.7%)	60 (74.1%)	69 (83.1%)
BMI, kg/m^2	25.3 (24.0-27.5)	25.3 (24.0-27.2)	25.3 (23.9-28.1)
Insulin dependent	50 (30.5%)	29 (35.8%)	21 (25.3%)
Hemoglobin A1c, %	7.3 (6.6-8.2)	7.4 (6.5-8.3)	7.3 (6.6-8.0)
Type of cardiac surgery			
On-pump CABG	154 (93.9%)	76 (93.8%)	78 (94.0%)
AVR	5 (3.0%)	2 (2.5%)	3 (3.6%)
CABG plus AVR	5 (3.0%)	3 (3.7%)	2 (2.4%)
CPB time, min	103 (80-121)	103 (83-125)	100 (77-120)





F Safety outcomes

Outcomes	Dapagliflozin (n=81)	Control (n=83)	*P*
Diabetic ketoacidosis	0	0	—
Urinary ketone after surgery			0.006
Trace (+)	8 (9.9%)	12 (14.5%)	
Positive (1+)	15 (18.5%)	11 (13.3%)	
Positive (2+)	12 (14.8%)	2 (2.4%)	
Positive (3+)	3 (3.7%)	0	
PH (POD 1-5)	7.39 (7.37-7.41)	7.40 (7.38-7.41)	0.045
HCO_3^- (POD 1-5)	23 (22-24)	24 (23-24)	<0.001
ICU readmission	1 (0.6%)	0	—
No cases of urogenital tract/incision infections, severe hypoglycemia or hypovolemia were observed in either group			
Follow-up at 30 days			
MACE	16 (19.8%)	22 (26.5%)	0.305
Myocardial infarction	14 (17.3%)	22 (26.5%)	0.154
Follow-up at 12 months			
MACE	16 (19.8%)	25 (30.1%)	0.125
Myocardial infarction	15 (18.5%)	23 (27.7%)	0.163

Figure 1 Overview of the DAPA-CS Trial. (*A*) Study design and population. Timeline of the intervention protocol. Dapagliflozin (10 mg/day) was administered preoperatively (Pre-op) for at least 3 days, withheld on the day of surgery, and resumed postoperatively (Post-op) upon oral intake until postoperative day (POD) 5. (*B*) CONSORT flow diagram. Patient screening, randomization, and analysis populations (mITT). (*C*) Baseline characteristics. Comparison of baseline data between groups. Data are presented as median (IQR), or number (%). (*D*) Myocardial injury. Temporal profile of hs-cTnI release. Data points represent geometric means with 95% CIs. The primary endpoint was the hs-cTnI level at 48 h postoperatively. AUC was calculated based on geometric means from 24 to 120 h after surgery. (*E*) Renal injury. Incidence of AKI (left) and temporal trajectories of serum creatinine (right). Data points represent geometric means with 95% CIs. AUC was calculated based on geometric means from 24 to 120 h after surgery. (*F*) Safety outcomes. Postoperative metabolic parameters and clinical adverse events. MACE was defined as a composite of all-cause mortality, myocardial infarction, stroke, and reoperation (at 30 days) or repeat revascularization/valve surgery (at 12 months). Myocardial infarction was defined according to the Fourth Universal Definition. AKI, acute kidney injury; AUC, area under the curve; AVR, aortic valve replacement; BMI, body mass index; CABG, coronary artery bypass grafting; CI, confidence interval; CPB, cardiopulmonary bypass; DKA, diabetic ketoacidosis; hs-cTnI, high-sensitivity cardiac troponin I; ICU, intensive care unit; IQR, interquartile range; MACE, major adverse cardiac events; mITT, modified intention-to-treat; POD, postoperative day; Post-op, postoperative; Pre-op, preoperative; SGLT2i, sodium-glucose cotransporter-2 inhibitor

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Statistical analysis

The sample size was based on the primary and key secondary end-points. A total of 162 patients were required to achieve 80% power (two-sided $\alpha = 0.05$) to detect a 30% reduction in 48-h hs-cTnI (log-transformed SD 0.35) and an effect size of 0.5 for AKI incidence (control incidence 40%). Allowing for an 8% dropout rate, 178 patients were enrolled. A prespecified hierarchical testing strategy controlled the overall type I error rate at 5%. Analyses were performed in the modified intention-to-treat (mITT) population, excluding randomized patients who did not undergo surgery.

Results

Between 2022 and 2024, 178 patients were randomized; 164 (81 dapagliflozin, 83 control) underwent surgery and were included in the mITT analysis (*Figure 1B*). Baseline characteristics were well balanced, with a median age of 64 years and 93.9% undergoing isolated on-pump CABG (*Figure 1C*).

For the primary end-point, hs-cTnI levels at 48 h were significantly lower with dapagliflozin (0.44 [0.26–1.16] vs 0.79 [0.43–1.50] ng/mL; $P = .004$) (*Figure 1D*). Additionally, the hs-cTnI area under the curve (AUC) (24–120 h) was reduced by 30% (48 vs 68 ng·h/mL; 95% CI, 5%–48%; $P < .001$), with consistently lower levels observed at other postoperative time points (24, 72, 96, and 120 h).

Regarding the key secondary end-point, AKI incidence did not differ between groups (40.7% vs 41.0%; RR 0.99; 95% CI, 0.69–1.44; $P = .977$)(*Figure 1E*), whereas the AUC for serum creatinine over 120 h was modestly but significantly reduced by 8% with dapagliflozin (6271 vs 6771 μmol/L·h; 95% CI, 2%–13%; $P = .013$).

Safety analyses showed that while the incidence of positive urine ketones was higher in the dapagliflozin group, no cases of DKA occurred (*Figure 1F*). Postoperative pH and bicarbonate levels were slightly lower in the dapagliflozin group, but remained within physiological ranges. No urogenital/incision infections, severe hypoglycaemia, or hypovolemia occurred in either group. There were no significant differences in 30-day or 1-year MACE; however, perioperative myocardial infarction (PMI) was numerically less frequent in the dapagliflozin group (*Figure 1F*).

Discussion

The DAPA-CS trial is the first to demonstrate that perioperative dapagliflozin reduces postoperative myocardial injury in cardiac surgery. The rapid cardioprotective effect may be attributed to multiple mechanisms. SGLT2i promote ketogenesis, and ketone bodies serve as an efficient energy substrate for the stressed myocardium, thereby preserving ATP homeostasis and attenuating inflammation during ischaemia–reperfusion.[8] Additionally, SGLT2i inhibit the sodium–hydrogen exchanger 1, a key mediator of reperfusion injury.[9] Concurrently, this biochemical cardioprotection was accompanied by a numerical reduction in PMI, suggesting that reducing subclinical injury may improve clinical outcomes.

While the incidence of AKI did not differ, lower serum creatinine release with dapagliflozin suggests an attenuation of renal stress. The limited sample size may have precluded detection of modest differences in AKI incidence. Moreover, the early afferent arteriolar vasoconstriction induced by SGLT2i causes a transient eGFR decline, potentially masking renoprotective effects in the acute phase; longer follow-up may be required to reveal sustained renal benefits.[2]

Importantly, our findings suggest that current conservative guidelines may warrant re-evaluation. Although recent evidence suggests a slightly increased postoperative euglycemic DKA risk with SGLT2i,[10] in our trial, withholding dapagliflozin on the day of surgery and resuming it after oral intake was re-established was not associated with DKA. The absence of DKA in our study may be explained by cardiac surgery–specific perioperative management. First, unlike gastrointestinal surgery, cardiac surgery patients typically resume oral intake within 24 h postoperatively, thereby reducing the metabolic drive for ketogenesis. Second, routine perioperative blood gas monitoring in cardiac surgery enables early detection and timely intervention for acid–base disturbances at subclinical stages. Third, our tailored dapagliflozin protocol, aligned with its pharmacokinetics and perioperative characteristics of cardiac surgery, was used to minimize DKA risk while preserving therapeutic benefits. Additionally, routine perioperative prophylactic antibiotics in cardiac surgery may reduce the risk of urogenital infections. Overall, SGLT2i demonstrate safety under appropriate perioperative monitoring and management. Cardiac surgery-specific perioperative practices may help optimize the risk–benefit balance of SGLT2i in this surgical setting.

Limitations include the single-centre, open-label design, potential Type I error from multiple secondary end-points, and the small sample size, which limits power to detect differences in hard clinical end-points. Nonetheless, these preliminary findings suggest that a tailored perioperative dapagliflozin strategy may be feasible and potentially cardioprotective, warranting further mechanistic investigation and validation in larger trials.

Acknowledgements

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Declarations

Disclosure of Interest

All authors declare no disclosure of interest for this contribution.

Data Availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Ethical Approval

This trial was approved by the institutional review board of Fuwai Hospital.

Pre-registered Clinical Trial Number

The pre-registered clinical trial number is NCT05621551.

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